

Image Processing Using OpenCV

1. Objective:

Apply basic image processing techniques using OpenCV (grayscale, blur, edge detection).

2. Tools Used:

- Python
- OpenCV
- Visual Studio Code

3. Description of Steps:

- Loaded the image using ``cv2.imread()``
- Converted it to grayscale using ``cv2.cvtColor()``
- Applied Gaussian Blur using ``cv2.GaussianBlur()``
- Detected edges using ``cv2.Canny()``

4. Output:

- Saved ``gray_image.jpg`` (Grayscale Image)
- Saved ``blurred_image.jpg`` (Blurred Image)
- Saved ``edges_image.jpg`` (Edge Detected Image)

5. Learning Outcome:

Gained hands-on experience with OpenCV and understood basic image manipulation steps.